# Vasily Ermilov

Embedded Software Engineer

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🔗 **github.com/miloVV13**

## 💼 Work Experience

### JSC «NIIME» — Embedded Software Engineer

2025 — present · Moscow, Zelenograd

- Embedded firmware development for STM32, ESP32, Raspberry Pi Pico
- FPGA logic design and digital interface implementation
- Low-level driver development in C/C++
- Test equipment integration via USB CDC/TMC/Ethernet with SCPI protocol
- Build system and version control: PlatformIO, CMake, Git

## 🎓 Education

### MIET National Research University — Bachelor's Degree

2022—2026

Department: Design and Manufacturing of Integrated Circuits
Direction: Electronics and nanoelectronics
Автоматизация проектирования изделий наноэлектроники
Period: 2022-2026
✅ Degree obtained

## 🛠 Technical Skills

**Languages** C C++ Python SQL HTML

**Platforms** STM32 ESP32 RP2040 FPGA ATmega328p

**Interfaces** USB CDC/TMC Ethernet UART/SPI/I2C SCPI

**Tools** Git PlatformIO CMake MatLab

## 🚀 Projects

### PC-to-Instrument Communication Gateway

C · USB CDC/TMC · Ethernet · SCPI · GUI

- Developed a gateway for controlling oscilloscopes and test equipment from PC
- Implemented SCPI protocol support for compatibility with professional software
- Built a graphical interface for data visualization and instrument control

### AI Assistant for Discord/Telegram Bots

Python · Local LLM · AsyncIO · Bot API

- Bot with local LLM integration for private, offline query processing
- Asynchronous architecture supporting multiple messaging platforms
- Data privacy focused: no cloud dependency

### MenuManager — GUI Library for Microcontrollers

C/C++ · PlatformIO · Cross-platform · WIP · HAL

- Library for creating menus on TFT/OLED/LCD displays
- Separation of logic and rendering for easy porting between platforms

🔗 **github.com/miloVV13/MenuManager**

## ℹ️ About Me

- **Qualities:** responsibility, systems thinking, quick learner
- **Interests:** embedded systems, measurement automation, photography
- **Languages:** Russian (native), English (technical)

Last updated: April 2026 · GitHub